

Designer Recommendations:

The current design functionally works. However, many improvements could be made. The main areas requiring improvements are the arm, tendon connection point, and cart connection point.

Arm Improvements:

The main improvements to the arm are in manufacturing. The main issues with the arm are weight and how it was cut. I would recommend using light-weight materials such as aluminum or PVC to solve the weight issue. However, you must ensure that the sections connecting to the cart are dense, as you want the material to stay upright while in use. The solution to fixing the manufacturing issue is to either use a CNC machine to cut off the sections where the arm connects to the hinge or create a new arm and handle that fits between the gap of the hinges; by doing so, you should be able to reduce the shifting the arm is observed doing.

Tendon Connection Point:

The main improvement to the tendon connection point is how it is made. The current issue is that I need to be more experienced in cutting the lumber into precise cubes. I recommend using a CNC to cut those connections or recruiting a well-trained wood worker to fix this issue.

Cart Connection Point:

The current setup using the clamps is quite a solid one. However, one could argue that they are unsightly, so I recommend creating your own clamps to fit your chosen appearance style. But if you are okay with the current looks of the clamps, they will work as intended.